

A required loop: of three policy/value routings in a recurrent vision-transformer model of covert attention, only the grounded, closed cortico–striatal–collicular loop learns

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Abstract

A behaving animal that attends must run two computations at once: a *policy*, deciding moment by moment whether a behaviourally relevant event has occurred, and a *value estimate*, predicting how good its current situation is — the quantity that, under any reward-driven account, supplies the teaching signal for the policy. We ask whether the way these two computations are *wired together* in a recurrent attention agent, and not merely their presence, determines whether attention-like behaviour can be learned at all. We study a conv-free recurrent vision-transformer agent trained purely from sparse reward on a spatially cued orientation-change-detection task ($T = 29$ frames) modelled on primate covert-attention paradigms. The agent carries two recurrent cross-attention streams: a *priority stream* (μ) that reads the live image and drives the actor (when to declare a change) and a *value stream* (q) that reads accumulated memory and drives a distributional critic. We compare three wirings that differ only in pathway coupling and readout routing. With a shared recurrent memory and the priority stream commanding the actor, the agent becomes a calibrated detector (accuracy 0.868, hit rate 0.794, false-alarm rate 0.067, reaction time ~ 1.95 frames) and reproduces the hallmark reliability-scaled cueing signature of primate spatial attention. Severing the coupling so each stream holds a private memory abolishes learning: the agent never once declares a change (hit rate 0.000, accuracy 0.504, the no-change base rate). A second control that retains the coupling but routes the value stream to the actor collapses identically (hit rate 0.000, accuracy 0.420). The effect is categorical — a ~ 79 -point hit-rate gap from a single structural edit — and establishes two jointly necessary conditions: the loop must be closed through a shared substrate, and the image-grounded stream must command the policy. We trace the failure to credit assignment: a critic that evaluates a different state than the actor occupies cannot teach it. We relate the working loop to the cortico-basal-ganglia–collicular circuit, presenting the correspondence as conceptual motivation rather than literal isomorphism, and use the trained model to generate falsifiable neurophysiological predictions.

1 Introduction

Selective visual attention requires a behaving animal to perform two logically distinct computations concurrently. It must form a *policy* — a moment-by-moment decision about what to do, such as whether to report that a behaviourally relevant event has occurred — and it must form a *value estimate*, a prediction of how good its current situation is, which under any reward-driven account is precisely the quantity that supplies the teaching signal for the policy. These

two computations draw on overlapping but distinguishable information. *When and where to act* is driven by bottom-up, image-grounded sensory evidence accumulated toward a commitment threshold (Gold and Shadlen, 2007); *how valuable the moment is* depends on accumulated context — which location was cued, how reliable the cue is, how much reward is at stake — a quantity carried on the same parietal substrate that supports the decision itself (Platt and Glimcher, 1999; Sugrue et al., 2004). This decomposition is the actor-critic organisation of reinforcement learning, and it has a deep counterpart in the biology: an actor for the policy and a critic for state value are carried by anatomically distinct but functionally interdependent circuits (O’Doherty et al., 2004; Maia, 2009).

The design question. Given an attention agent that must compute both a policy and a value, how should the two computations be wired? A natural engineering instinct is to *separate* them: give each its own specialised recurrent stream so that the image-reading machinery that decides when to act is insulated from the context-integrating machinery that estimates value. Separation is attractive on grounds of modularity and interpretability, and for purely feedforward representations there is evidence that it helps. The alternative is to *couple* the streams through a shared, jointly maintained working memory, so that the actor writes the very state the critic evaluates. Which organisation is required — and whether the question even has a categorical answer — is the subject of this paper.

A lesion-style controlled comparison. We train a conv-free recurrent vision-transformer agent purely from sparse reward on a spatially cued orientation-change-detection task ($T = 29$ frames) built to mirror primate covert-attention paradigms. The agent embeds each 50×50 frame as 100 patch tokens on a 10×10 grid, maintains per-token recurrent memories through a cross-attention encoder, and reads out two heads: an actor that emits a binary wait/declare action each frame and a distributional (quantile) critic. Two recurrent streams feed these heads — a *priority stream* (μ) whose residual preserves the raw image and which therefore carries live visual evidence to the actor, and a *value stream* (q) whose residual is the deep memory and which carries accumulated context to the critic. (In an older vocabulary these are the “salience” and “top-down” streams; we use priority and value throughout.) We then hold the task, the learning algorithm, the hyperparameters and the number of updates fixed and vary a single structural property across three configurations, in the manner of a lesion experiment. The first *couple*s the streams through a shared recurrent memory with the priority stream commanding the actor. The second *sever*s the coupling — each stream is given a private memory updated only by its own output, the minimal edit that removes cross-talk while preserving the split, the stream types, the parameter count and the recipe. The third *retains* the coupling but *swaps the routing*, so the value stream drives the actor and the priority stream drives the critic. Only one wiring can show what is structurally required, because the other two isolate the two candidate necessary conditions.

Preview of the result. The outcome is categorical rather than graded. The coupled, correctly routed agent becomes a calibrated psychophysical detector (accuracy 0.868, hit rate 0.794, false-alarm rate 0.067, reaction time ~ 1.95 frames) that reproduces the reliability-scaled validity benefit of primate spatial attention. Severing the coupling collapses the agent to always-wait: across 500 episodes it never once declares a change (hit rate 0.000; accuracy 0.504, exactly the no-change base rate). Swapping the routing collapses it identically (hit rate 0.000; accuracy 0.420). Learning therefore requires *both* that the loop be closed through a shared recurrent memory *and* that the grounded priority stream command the policy; a single structural edit to either condition produces a ~ 79 -point hit-rate gap. We trace the collapses to reinforcement-learning credit assignment — a critic that evaluates a state the actor does not occupy cannot

teach it, and an actor fed a representation that cannot see the change has nothing to which credit can be assigned.

Brain-loop interpretation and novelty. We relate the one working configuration to the primate cortico-basal-ganglia-collicular loop: a grounded priority and commitment node (visual cortex, FEF-visual cells, superior colliculus) that commits the action and writes the shared working state (Thompson et al., 2005; Moore and Fallah, 2001; Lovejoy and Krauzlis, 2010; Zénon and Krauzlis, 2012); a value/critic system in the basal ganglia that reads that state to evaluate it (O’Doherty et al., 2004; Maia, 2009); and a dopaminergic reward-prediction-error signal from substantia nigra pars compacta that closes the loop by teaching the actor’s plasticity (Schultz et al., 1997; Hollerman and Schultz, 1998; Morita and Kato, 2014). We present this correspondence as conceptual motivation, not literal isomorphism, and state its limits plainly in the Discussion. The result is, to our knowledge, novel for a recurrent, image-grounded agent: no prior model splits a recurrent vision encoder into priority and value streams on a shared memory, requires that coupling for learning, or requires the grounded stream to be the actor. It stands in clean contrast to the one machine-learning finding on actor/critic representation sharing, which reports the *opposite* for feedforward representations — there, separation helps. For a recurrent agent that must read a live image to act, coupling is required.

2 The three loop architectures

The agent is a conv-free recurrent vision transformer. Each $50 \times 50 \times 3$ frame $\mathbf{O}^{(t)}$ is partitioned into a 10×10 grid of non-overlapping 5×5 patches, yielding $N = 100$ tokens $\mathbf{X}^{(t)} = \{\mathbf{x}_i^{(t)}\}_{i=1}^N$, each embedded by a shared per-patch network into a $d = 128$ -dimensional vector and tagged with learned positional and temporal codes. Information passes between spatial locations only through attention. Two per-token recurrent memories are carried across frames as hidden states, one slot per patch location, $\mathbf{H}_1^{(t)}, \mathbf{H}_2^{(t)} \in \mathbb{R}^{N \times d}$, advanced by independent long short-term memory (LSTM) cells. We refer to $\mathbf{H}_1$ as the sensory trace, written directly from the raw image, and to $\mathbf{H}_2$ as the deep trace, written from the decision pathway’s own output. The encoder is split into two cross-attention streams that read these memories: a *priority* stream (μ), which is grounded in the live image and drives the ACTOR (policy), and a *value* stream (q), which is grounded in accumulated memory and drives the CRITIC (valuation). In the older salience/top-down vocabulary, the priority stream is the salience pathway and the value stream the top-down pathway; we use priority and value throughout.

2.1 The cross-attention block

Each stream is a single-head, pre-norm cross-attention block whose *queries* are the current image tokens and whose *keys and values* are supplied by memory. Writing $\text{Attn}(Q; K) = \text{softmax}(QK^\top/\sqrt{d})V$ with V tied to K , a residual source R , and a position-wise feed-forward block FFN,

$$\mathbf{Z} = R + \text{Attn}(\mathbf{X}; \text{KV}), \quad \mathbf{Z} \leftarrow \mathbf{Z} + \text{FFN}(\mathbf{Z}). \quad (1)$$

The two streams differ only in their key/value bank KV and residual R .

2.2 Configuration 1 (V1): shared memory, closed loop

In the working configuration the two streams are coupled through a single shared deep memory $\mathbf{H}_2$:

$$\text{priority } (\mu, \rightarrow \text{ACTOR}) : \quad \mathbf{Z}_\mu^{(t)} = \text{block}(\mathbf{X}^{(t)}; \text{KV} = [\mathbf{H}_1^{(t)} \parallel \mathbf{H}_2^{(t)}], R = \mathbf{X}^{(t)}), \quad (2)$$

$$\text{value } (q, \rightarrow \text{CRITIC}) : \quad \mathbf{Z}_q^{(t)} = \text{block}(\mathbf{X}^{(t)}; \text{KV} = \mathbf{H}_2^{(t)}, R = \mathbf{H}_2^{(t)}). \quad (3)$$

The priority stream attends over *both* memories, concatenated along the token axis with each source tagged, and carries the raw image on its residual; its decision can therefore reflect live visual evidence. The value stream attends only over the deep memory $\mathbf{H}_2$ and carries $\mathbf{H}_2$ on its residual, so the image can only *re-weight* the stored state rather than inject new content—a memory-grounded bottleneck appropriate to a contextual value estimate. The memories are then advanced,

$$\mathbf{H}_1^{(t+1)} = \text{LSTM}_1(\mathbf{X}^{(t)}, \mathbf{H}_1^{(t)}), \quad \mathbf{H}_2^{(t+1)} = \text{LSTM}_2(\mathbf{Z}_\mu^{(t)}, \mathbf{H}_2^{(t)}), \quad (4)$$

so the sensory trace is written from the raw image and the deep trace is written from the priority/actor output $\mathbf{Z}_\mu$. Finally a split readout sends each stream to its own head:

$$\pi^{(t)} = \text{ACTOR}(\mathbf{Z}_\mu^{(t)}), \quad Q^{(t)} = \text{CRITIC}(\mathbf{Z}_q^{(t)}). \quad (5)$$

The coupling lives entirely in $\mathbf{H}_2$: the priority/actor stream *writes* $\mathbf{H}_2$ (through $\mathbf{Z}_\mu$ in Equation (4)) and *reads* it back (inside the concatenated bank of Equation (2)), while the value/critic stream *reads* the same $\mathbf{H}_2$ (Equation (3)). The two heads are separate but sit on one shared, jointly maintained working memory—a common workspace that the decision system writes and both systems read. This shared register is the structural variable we manipulate.

2.3 Configuration 2 (V2): independent streams, open loop

The second configuration keeps both stream *types* but gives each its own private memory, updated only by its own output:

$$\mathbf{Z}_\mu^{(t)} = \text{block}(\mathbf{X}^{(t)}; \text{KV} = \mathbf{H}_\mu^{(t)}, R = \mathbf{X}^{(t)}), \quad \mathbf{H}_\mu^{(t+1)} = \text{LSTM}_\mu(\mathbf{Z}_\mu^{(t)}, \mathbf{H}_\mu^{(t)}), \quad (6)$$

$$\mathbf{Z}_q^{(t)} = \text{block}(\mathbf{X}^{(t)}; \text{KV} = \mathbf{H}_Q^{(t)}, R = \mathbf{H}_Q^{(t)}), \quad \mathbf{H}_Q^{(t+1)} = \text{LSTM}_q(\mathbf{Z}_q^{(t)}, \mathbf{H}_Q^{(t)}). \quad (7)$$

The priority module still drives the actor and the value module the critic, but the two modules never touch each other’s recurrent state; the only thing they share is the raw input image. This is the minimal edit that removes the cross-talk while preserving the split, the stream types, the parameter count and the training recipe. The loop is open: nothing the actor writes is read by the critic, and nothing the critic computes flows into the actor’s representation.

2.4 Configuration 3 (V3): swapped routing

The third configuration is architecturally identical to V1—the same shared $\mathbf{H}_2$, the same memory writes of Equation (4), the same two streams of Equations (2)–(3)—and changes only the readout of Equation (5): the value/memory stream drives the actor and the priority/image stream drives the critic,

$$\pi^{(t)} = \text{ACTOR}(\mathbf{Z}_q^{(t)}), \quad Q^{(t)} = \text{CRITIC}(\mathbf{Z}_\mu^{(t)}). \quad (8)$$

The loop remains closed; only the assignment of streams to heads is reversed. This isolates whether it matters *which* stream commands the policy, holding the presence of coupling fixed.

2.5 One property at a time

The three configurations differ in exactly one property each, relative to the working loop. V2 removes the coupling (private memories) while preserving the split, the stream types, the routing, the parameter count and the recipe. V3 preserves the coupling and reverses only the routing. Everything else—patch front-end, two LSTM memories, single-head pre-norm blocks, both heads, the optimiser, hyperparameters and number of updates—is held identical across all three. The comparison therefore isolates the causal contribution of *organisation*, in the spirit of a reversible inactivation experiment (Lovejoy and Krauzlis, 2010).

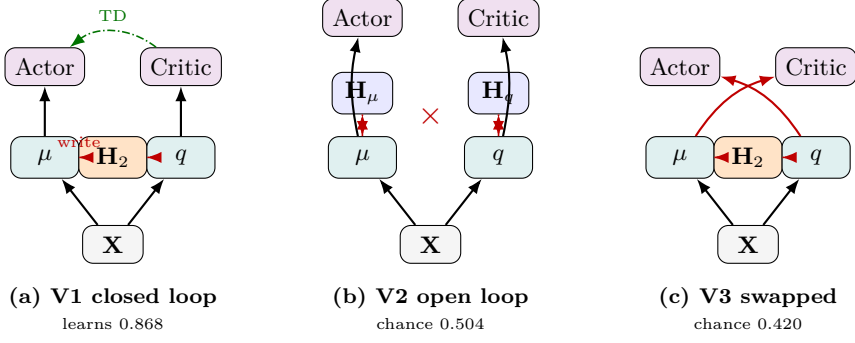


Figure 1. The three loop architectures. **V1 (shared memory, closed loop)**: the priority stream (μ ; queries the image, keys/values $[\mathbf{H}_1 \parallel \mathbf{H}_2]$, residual the raw image $\mathbf{X}$) drives the actor and *writes* the shared deep memory $\mathbf{H}_2$; the value stream (q ; keys/values and residual $\mathbf{H}_2$) drives the critic and *reads* the same $\mathbf{H}_2$. **V2 (independent, open loop)**: the same two stream types, but each with a private memory ($\mathbf{H}_\mu, \mathbf{H}_q$) written only by its own output; the streams share only the raw image. **V3 (swapped routing)**: identical wiring to V1 but the readout is reversed—the value stream drives the actor and the priority stream drives the critic. The three configurations differ in exactly one property each: V2 deletes the shared memory, V3 swaps the head assignment.

3 Task and training

3.1 The cued change-detection environment ($T = 29$)

The agent performs a spatially cued orientation-change-detection task over an episode of $T = 29$ frames. Each 50×50 frame is divided into the same 10×10 grid of 100 patch tokens the encoder reads. On an early frame a coloured cue indicates one side of the display and is then removed. The cue carries two pieces of information. Its *reliability*—the validity, drawn from $\{0.25, 0.50, 0.75, 1.00\}$ —sets the probability that an upcoming orientation change occurs at the cued location; its *colour* sets the reward magnitude available on that trial, so the cue specifies both where to attend and how much the trial is worth. Half of all trials are change trials. On a change trial the orientation of the target patch shifts by a magnitude of up to roughly 64° , with the change onset jittered across frames (the change index falls in roughly $[11, 25]$) and the new orientation persisting to the end of the episode; the remaining trials contain no change. Frame-to-frame Gaussian orientation noise ($\sigma = 5$) sets the discrimination difficulty, so that small changes must be detected against ongoing sensory fluctuation. On every frame the agent emits a binary action, *wait* or *declare change*. Declaring at or after the true change onset is a hit; declaring before it is a false alarm; waiting out a no-change trial to the end is a correct rejection. The long horizon, the jittered onset, the removed-then-remembered cue, and the graded reliability together make this a test of maintained covert spatial attention rather than of a single-frame discrimination, and the cue-colour reward structure adds a value dimension on top of the spatial one.

3.2 The shared training harness

All three configurations are trained by an identical offline actor–critic reinforcement-learning procedure, with the same hyperparameters and the same number of updates; the configurations differ only in the encoder wiring of Section 2. The actor and a distributional critic are one-dimensional-convolutional heads over the token axis: the actor emits the wait/declare logit pair and the critic estimates a five-quantile distribution over returns. The actor loss is a maximum-a-posteriori policy-improvement (MPO/MAP) term blended with a behaviour-cloning term; the critic is trained with a quantile (distributional) Bellman loss against a periodically hard-copied target network that supplies both the critic’s bootstrap target and the policy-improvement

Table 1. Behavioural outcomes of the three loop architectures on the $T = 29$ cued change-detection task (500 fresh episodes each, sampled policy). The shared-memory closed loop (V1) is a calibrated, change-locked detector; the independent (V2) and swapped-routing (V3) configurations both collapse to never declaring a change. Each configuration differs from V1 in exactly one structural property.

Configuration	Loop	Accuracy	Hit rate	FA rate	RT (frames)
V1 shared memory	closed, $\mu \rightarrow$ actor	0.868	0.794	0.067	1.95
V2 independent	open	0.504	0.000	0.000	—
V3 swapped routing	closed, $q \rightarrow$ actor	0.420	0.000	0.000	—

reference. Whole episodes are stored in a prioritised replay buffer, with priorities set by per-episode quantile error, and a brief forced-wait warm-up trains the critic on full trajectories before the actor begins to act, so the policy is first improved against an already-informative value estimate. Because the harness, the parameter budget and the update count are held fixed, any difference in outcome across the three configurations is attributable to the single structural property that distinguishes them.

4 Results

4.1 Only the closed, grounded loop learns

We trained each of the three configurations on the cued change-detection task with identical task, algorithm, hyperparameters and number of updates, and evaluated each on 500 fresh episodes under a stochastic (sampled) policy. Outcomes were scored by signal-detection category—hit, miss, false alarm, correct rejection—and summarised by overall accuracy, hit rate on change trials, false-alarm (FA) rate on no-change trials, and reaction time (RT) on hits (Table 1).

The shared-memory closed loop (V1) is a genuine psychophysical detector. It catches roughly four of every five real changes (hit rate 0.794), rarely fires on no-change trials (FA = 0.067), and responds quickly—a mean reaction time of 1.95 frames after change onset—for an overall accuracy of 0.868. The other two configurations do not partially learn the task; across 500 episodes each *never once* declares a change. Their hit and FA rates are both exactly 0.000. The independent configuration’s accuracy, 0.504, is precisely the rate obtained by waiting out every trial and being correct only on the half that contain no change; the swapped-routing configuration reaches 0.420 by the same always-wait strategy. The policy has retreated, in both cases, to the single safe action—always wait—that can never incur a false-alarm penalty, and it never discovers that declaring a change is worthwhile.

The effect is categorical, not a matter of degree. A single structural edit—removing the shared memory (V1→V2) or reversing which stream commands the actor (V1→V3)—turns a 0.794 hit rate into 0.000, a 79.4-percentage-point gap, and turns a 0.868 accuracy into chance-level always-wait. Two conditions are jointly necessary for the task to be learned at all: the loop must be *closed* through a shared recurrent memory (V2 fails without it), and the *grounded priority stream must command the policy* (V3 fails when the value stream does). Only V1 satisfies both.

4.2 The working loop reproduces the reliability-scaled cueing signature

A non-collapsed accuracy number is necessary but not sufficient evidence that the right computation has emerged. The discriminating test is whether the agent shows the behavioural fingerprint of covert spatial attention: a benefit for changes at the cued location that scales with how reliable the cue is. Detailed characterisation of V1 confirms this signature.

Splitting hit rate by cue reliability reveals a graded cueing effect that rises monotonically with reliability: 0.680 at validity 0.25, 0.697 at 0.50, 0.750 at 0.75, and 0.753 at 1.00 (each

cell $n = 300$; sampled policy). Decisions remain change-locked at every level, with a median reaction time of 3.0 frames at all four reliabilities and a premature-press rate at or below 0.0033. The advantage of a change at the cued (valid) location over an equivalent change at an uncued (invalid) location is largest where the psychometric function is steepest—near a change magnitude of 14° —and it grows with reliability: the valid–invalid hit-rate gap is +0.092 at validity 0.25 (0.748 vs. 0.656), +0.132 at 0.50 (0.756 vs. 0.624), +0.136 at 0.75 (0.736 vs. 0.600), and +0.208 at 1.00 (0.776 vs. 0.568). At ceiling and floor magnitudes the gap closes, as expected for an effect expressed as a shift of perceptual threshold rather than a uniform additive benefit. The underlying psychometric function is a clean orientation-discrimination sigmoid: at validity 1.00 the valid-location hit rate runs from 0.016 at 2° through 0.308 (9°), 0.776 (14°), 0.976 (20°) to 1.000 at 30° and above (each cell $n = 250$). Reaction time on valid trials likewise tracks difficulty: at the lowest reliability (validity 0.25) the median falls from 10.0 frames at 2° to 4.0 (9°), 3.0 (14°), 2.0 (20 – 30°) and 1.0 frame at 44° , with the same monotone speed-up at every reliability. The model thus reproduces the canonical covert-attention pattern—a graded, reliability-scaled validity benefit expressed as a lowered threshold for the attended location—in an agent trained only to detect changes for reward.

The two collapsed configurations, by contrast, have no behavioural signature to characterise. An agent that never presses produces a flat zero hit rate at every reliability and validity condition; there is no cueing benefit, no psychometric function, and no reaction-time structure to fit. The capacity to express spatial attention is downstream of the capacity to act at all, and only the closed, grounded loop clears that first bar.

5 Why only the closed, grounded loop learns

The categorical gap of Section 4 follows from how actor–critic reinforcement learning assigns credit, and from what each stream can and cannot see. The actor improves its policy by following an advantage—roughly, “is acting now better than the critic’s estimate of the current state’s value?”—and that signal is only useful if the critic is evaluating the same state the actor is acting from, and if the actor’s input actually contains the evidence the action should depend on. The three configurations break, preserve, or misroute exactly these requirements.

V1: credit lands where it was earned. In the closed, grounded loop the actor and critic operate over one shared deep memory $\mathbf{H}_2$, and the actor’s own output writes that memory (Equation (4)). The critic therefore evaluates the same evolving state the actor occupies, and “pressing now looks good” is a statement about the state the actor actually inhabits. Credit lands on the representation that produced the action. Because the priority stream carries the raw image on its residual and reads both memories, the actor’s input contains live visual evidence of the orientation change. The agent can therefore discover that declaring at change onset is rewarded, and can shape that policy by cue reliability and value—the covert-attention signature of Section 4 emerges as the learned solution.

V2: credit is computed on the wrong representation. When the streams are given private memories, the actor acts from $\mathbf{H}_\mu$ while the critic values $\mathbf{H}_Q$, and the two recurrent states drift apart over the episode. The advantage handed to the actor is then computed on a representation that is not the one the actor is acting from, so credit is systematically misattributed: an action that helped may be evaluated against a value estimate for a different internal state. Under such mis-specified feedback the policy falls into the only basin that is safe under *any* value estimate—never act, never be penalised—and stays there. The always-wait collapse is the rational endpoint of an agent that cannot tell when acting helps. It is a learning failure, not a capacity failure: the network has the parameters and the pathways to express

the task (V1 proves it), but with the loop open the teaching signal never reaches the right representation.

V3: the actor is fed a representation that cannot see the change. Swapped routing preserves the coupling—value and policy are still bound through the shared $\mathbf{H}_2$, and credit assignment is clean—but the actor is now driven by the value stream, whose keys, values and residual are all the deep memory $\mathbf{H}_2$ rather than the image. That stream carries almost no live visual content: the image can only re-weight stored state, not inject new evidence. The decision is therefore made from a representation that cannot *see* the orientation change as it happens. However well-posed the credit assignment, there is nothing change-related in the actor’s input to assign credit to, and the policy again retreats to always-wait. V3 thus dissociates the two requirements: coupling alone is not enough; the stream that commands the action must be the grounded one.

Jointly necessary, individually insufficient. The three outcomes form a clean logical decomposition. V2 has a grounded actor but an open loop and fails. V3 has a closed loop but a non-grounded actor and fails. Only V1 has both a closed loop and a grounded actor, and only V1 learns. Learning the task requires that the policy and value computations be coupled through a shared recurrent state *and* that the image-grounded priority stream be the one that commands the policy—two conditions that a single structural edit can each remove, and whose removal is catastrophic rather than graded.

6 A cortico-basal-ganglia-collicular loop

The controlled comparison isolates two structural conditions that are jointly necessary for learning: the policy and value computations must be coupled through a shared, jointly-maintained recurrent memory (V2 fails without it), and the grounded *priority* stream must command the actor (V3 fails when the *value* stream does instead). Neither condition is an arbitrary engineering choice. Each maps onto a well-characterised feature of the cortico-basal-ganglia-collicular machinery that the primate brain uses to convert sensory evidence into committed, value-weighted action. We develop that mapping here, both because it motivates the architecture and because it makes the two failure modes biologically interpretable rather than incidental.

6.1 Actor and critic as a striatal division of labour

The model factors the decision into an actor that emits the wait/declare policy and a distributional critic that evaluates the state. This factorisation has a direct anatomical precedent. In the basal ganglia, the dorsal striatum is most closely associated with the actor—the mapping from state to action—while the ventral striatum, together with its dopaminergic input, is most closely associated with the critic, the evaluation of state value and the computation of a prediction error (O’Doherty et al., 2004; Maia, 2009). Human imaging during instrumental learning dissociates these roles: ventral striatal activity tracks reward prediction in both Pavlovian and instrumental settings, whereas dorsal striatal activity is selective for the instrumental, action-contingent case (O’Doherty et al., 2004). The two systems are anatomically distinct but functionally interdependent, which is exactly the relationship the model instantiates between its separate actor and critic heads sitting on one shared working memory.

We stress that this correspondence is an organising idealisation rather than settled anatomy. A literal localisation of “the critic” to a single nucleus overstates the cleanliness of the division: action-value and reward signals are distributed broadly across the striatum, and the dorsal/ventral split is better read as a gradient than a partition (Maia, 2009). We therefore use the actor-critic mapping as a conceptual frame for the model’s labour division, not as a claim

that the priority stream μ and the value stream q are two anatomically separable attention nodes.

6.2 Dopamine as the coupling that makes V2’s collapse sensible

The sharpest empirical result is that severing the coupling between the policy and value computations abolishes learning entirely: the independent variant (V2), in which the actor acts from a private memory and the critic evaluates a separate private memory, collapses to an always-wait policy with a hit rate of 0.000, never once declaring a change across 500 fresh episodes. This is not a capacity failure—the network retains the parameters and pathways to express the task—but a credit-assignment failure. When the actor and critic occupy different representations, the advantage handed to the actor is computed on a state the actor does not occupy, credit is systematically misattributed, and the policy settles into the only basin that is safe under any value estimate: never act, and never be penalised.

The biological counterpart of the required coupling is the dopaminergic teaching signal. Mid-brain dopamine neurons in the substantia nigra pars compacta (SNc) and the ventral tegmental area encode a temporal-difference reward-prediction error: they respond to unexpected reward, shift their response to the earliest reward-predicting cue as learning proceeds, and are suppressed when a predicted reward is omitted (Schultz et al., 1997; Hollerman and Schultz, 1998). This error is precisely the quantity the critic computes, and it is broadcast to striatal synapses where it gates plasticity. The corticostriatal learning rule is a three-factor rule—presynaptic activity, postsynaptic activity, and dopamine concentration jointly determine whether a corticostriatal synapse potentiates or depresses—so that the policy encoded in striatal weights is updated only in proportion to the dopaminergic error supplied by the value system (Morita and Kato, 2014). Value and policy are thereby bound into a single closed loop: the actor learns nothing except through the critic’s broadcast error.

This is the conceptual precedent for V2. An agent whose policy circuit receives no teaching signal derived from the same evaluated state has no instructive gradient with which to discover that declaring at change onset is rewarded; the always-wait collapse is the rational endpoint of an actor that cannot tell when acting helps. In the model, the shared deep memory H_2 plays the role the dopaminergic loop plays in the brain: because the actor’s own output writes H_2 and the critic reads the same H_2 , “pressing now looks good” is a statement about the state the actor actually occupies, and credit lands where it was earned. Removing the shared memory is the in-silico analogue of opening the dopaminergic teaching loop.

The parallel must be drawn with care, and we flag the principal disanalogy explicitly. In the brain the critic-to-actor influence is a *broadcast scalar*—a diffusely released dopamine prediction error flowing from the critic’s evaluation back to the actor’s synapses. In the model the coupling is a *shared recurrent state* that the actor writes and both streams read. The substrate and the dominant direction of the brain’s coupling (critic $\rightarrow$ actor, via dopamine) differ from the model’s most visible coupling (priority $\rightarrow H_2$, a written state). We present the correspondence as motivation, not isomorphism. We note, however, that both directions genuinely co-exist in the model: the priority stream’s write into H_2 closes the loop forward, while the critic’s gradient—the model’s analogue of the dopamine error—closes it backward by teaching the actor. It is, in both systems, a loop; what we decline to claim is that the loop is realised by the same machinery.

We further note that some of the strongest evidence that value and decision signals can share a substrate comes from parietal cortex: neurons in the lateral intraparietal area (LIP) carry both the accumulating decision variable and reward-magnitude or reward-probability signals on the same population, with reward entering as an offset to the decision process (Platt and Glimcher, 1999; Sugrue et al., 2004). That a single neural substrate intermixes value and decision lends biological plausibility to the model’s premise that one shared working memory can support both an actor and a critic read-out.

6.3 The superior colliculus as the grounded node that must command action

The second necessary condition—revealed by the swapped variant (V3), which is architecturally identical to the working model but routes the value stream to the actor and the priority stream to the critic, and which likewise collapses to a hit rate of 0.000—is that the actor must be driven by the image-grounded stream. In V3 the loop is closed and credit assignment is clean, yet the actor is fed a representation whose residual is memory rather than the live image; it carries almost no current visual content, so there is nothing in the actor’s input from which to detect the change. The grounded stream is what the policy must read.

This condition maps onto the role of the superior colliculus (SC), and of the visual cells of the frontal eye field (FEF), as the grounded priority node that commits the response. The SC is causally necessary for covert spatial attention: reversible inactivation of the intermediate SC produces a marked deficit in covert target selection even when the eyes never move, and does so without changing the attentional gain measured in extrastriate cortex (Zénon and Krauzlis, 2012; Lovejoy and Krauzlis, 2010). The colliculus is best understood not as a gain knob on sensory cortex but as the threshold that converts accumulating cortical evidence into a committed choice: collicular activity reflects the commitment to a response rather than the evidence accumulation itself (Stine et al., 2023). Its causal weight is heavy—SC inactivation is more devastating to covert attentional performance than FEF inactivation (Bollimunta et al., 2018)—and its influence extends into the basal ganglia: attentional modulation in the caudate nucleus depends on an intact SC, defining an SC-caudate-cortex loop in which the collicular priority signal conditions striatal processing (Herman et al., 2020). That loop is a loose biological analogue of the model’s arrangement, in which the priority stream both commits the action and writes the shared state that the value circuit then reads.

The complementary cortical evidence points the same way. During covert attention the FEF carries the attentional signal on its *visual* (image-grounded) cells, while its movement-related cells are suppressed—the grounded representation, not the motor command, does the perceptual work (Thompson et al., 2005). Subthreshold microstimulation of the FEF, below the level that evokes a saccade, enhances visual sensitivity at the corresponding location, establishing the FEF as a controller that acts through the grounded visual pathway (Moore and Fallah, 2001, 2004). More generally, perceptual decisions are formed by integrating grounded sensory evidence to a bound (Gold and Shadlen, 2007), and the grounded representation is the indispensable input: inactivating sensory cortex (MT) impairs the discrimination itself, whereas inactivating the downstream accumulator (LIP) leaves the discrimination largely intact (Katz et al., 2016). This dissociation is the single closest biological parallel to V3—an actor cut off from the grounded sensory representation cannot perform the discrimination no matter how well-formed its downstream evaluation.

We again temper the mapping. The model’s priority stream μ corresponds most naturally to the FEF-visual and SC commitment integrator, and the value stream q to the value/critic system in the basal ganglia and its dopaminergic input; the latter is not itself an attention node. SC, FEF, and LIP are each single priority maps, not actor-critic pairs, and we do not claim that μ and q map onto two distinct attention areas. The claim is narrower and, we think, robust: a grounded, collicular-style commitment node must be the one that drives the action, and a value circuit must be the one that evaluates it.

6.4 The loop, stated whole

The three results assemble into a single closed loop. Grounded priority—visual cortex feeding the SC and the FEF-visual cells—commits the declare response and writes the shared working state. The value system in the striatum and basal ganglia reads that state and evaluates it. The evaluation is broadcast back as an SNc dopamine temporal-difference error that teaches the actor’s plasticity, closing the loop. Only when the loop is closed (a shared state coupling

policy and value) *and* the grounded node commands the action does the system learn. V2 corresponds to an open loop: the policy circuit receives no teaching signal tied to the evaluated state, as if the dopaminergic coupling reaching the actor were severed. V3 corresponds to a loop in which the value circuit wrongly issues the motor commitment without grounded collicular evidence—credit assignment is intact, but the commanding representation cannot see the world. The working model is the one configuration that respects both the closure and the grounding of the biological circuit, and it is the only one of the three that learns the task.

7 Falsifiable predictions

The loop hypothesis makes a concrete, falsifiable commitment in the form of a double dissociation. The two streams carry functionally distinct information—the priority stream the grounded change-evidence that drives detection, the value stream the cue-value and reliability context that modulates it—so selectively disabling each should produce opposite, separable behavioural signatures. Lesioning the value circuit should degrade value and context modulation while sparing change detection; lesioning the priority node should collapse detection itself. We state the prediction at three levels of measurement, each independently testable, and tie each to a corresponding manipulation in the model and in the animal.

7.1 Behavioural level

The coarsest test is the one the present comparison already exemplifies and which can be reproduced as an intervention rather than an architectural ablation. Two behavioural laws follow from the loop hypothesis. First, *coupling is required*: disrupting the shared recurrent state—so that the actor and critic no longer occupy a common representation—should abolish learning of the declare response, reproducing the always-wait collapse seen in the independent variant (hit rate 0.000 versus the working model’s 0.794). Second, *the grounded stream must command the action*: routing the policy read-out from the value stream rather than the priority stream should likewise collapse detection, as in the swapped variant. Both are categorical, all-or-nothing effects—a hit-rate gap of roughly 79 points from a single structural edit—rather than graded performance changes, which makes them stringent tests: the hypothesis forbids a working model that violates either condition.

The corresponding animal prediction is a dissociation between two manipulations. Inactivating the priority/commitment node—the superior colliculus—should collapse covert change detection, mirroring the loss of the grounded actor input; this is consistent with the established finding that SC inactivation produces a large covert-attention deficit (Zénon and Krauzlis, 2012; Lovejoy and Krauzlis, 2010; Bollimunta et al., 2018), and the prediction sharpens it to the orientation-change-detection setting and to optogenetically precise, reversible perturbation (Krauzlis et al., 2018; Katz et al., 2025). By contrast, a manipulation confined to the value circuit—the dopaminergic teaching system or its ventral-striatal target—should leave the animal’s ability to detect a change broadly intact while degrading how strongly cue value and cue reliability modulate behaviour. In the model the working variant exhibits exactly the modulation that such a value lesion should target: a graded, reliability-scaled validity benefit, with per-validity hit rates rising monotonically from 0.680 at validity 0.25 through 0.697 and 0.750 to 0.753 at validity 1.00, and, at matched change magnitude, a valid-over-invalid advantage that scales with reliability (for example at a 14° change, 0.748 valid versus 0.656 invalid at validity 0.25, widening to 0.776 versus 0.568 at validity 1.00). The double dissociation predicts that a priority/SC lesion flattens detection regardless of these contextual variables, whereas a value lesion flattens the reliability-scaled benefit while detection of large, salient changes survives.

7.2 Representational level

The streams should be *decodably* specialised. The priority stream’s output Z_{pri} , whose residual preserves the raw image, should carry the change-evidence: a decoder trained on Z_{pri} should read out the presence and location of the orientation change far better than one trained on the value stream’s output Z_{val} , whose residual is the deep memory and which carries little live visual content. Conversely, the value/context variables—cue value (reward magnitude) and cue reliability (validity)—should decode far better from Z_{val} than from Z_{pri} . This is a within-model prediction that is directly verifiable from the activations and that grounds the lesion logic: it asserts that the two streams are not redundant copies but information-segregated channels, so that disabling one removes a specific, identifiable variable. The animal analogue is the expectation that grounded sensory and commitment areas (sensory cortex, FEF-visual, SC) carry the discriminative change-evidence (Thompson et al., 2005; Gold and Shadlen, 2007; Katz et al., 2016), whereas the striatal/value system carries the reward and probability context on a partly shared substrate (Platt and Glimcher, 1999; Sugrue et al., 2004).

7.3 Causal-perturbation level

The strongest test is causal and graded rather than ablative. Perturbing the priority attention computation—scaling or clamping the grounded stream that feeds the actor—should shift behaviour and the detection threshold: a weakened priority signal should raise the effective change magnitude needed to declare, lengthen reaction times (which in the intact model fall monotonically with change magnitude, from a median of 10 frames at 2° to 1 frame at 44°), and ultimately abolish detection as the perturbation deepens. Perturbing the value attention computation—clamping the stream that feeds the critic—should instead shift the critic’s value estimates and the cue-value and reliability modulation while leaving the detection threshold and reaction-time structure for large changes essentially unchanged. This is the model-internal realisation of the double dissociation: priority perturbation moves the *behavioural threshold*, value perturbation moves the *critic’s valuation*, and the two effects are separable.

The matched animal experiment is graded optogenetic or microstimulation control of the superior colliculus versus a manipulation of the dopaminergic value circuit. Subthreshold collicular and frontal manipulations are known to shift covert attentional performance in a graded fashion (Moore and Fallah, 2001, 2004; Stine et al., 2023; Herman et al., 2020), and recent optogenetic SC perturbation provides the precise, reversible tool the prediction requires (Katz et al., 2025; Krauzlis et al., 2018): graded SC suppression should produce a graded rise in detection threshold up to collapse, whereas a value-circuit manipulation of comparable behavioural scale should leave detection of salient changes intact while selectively blunting the reliability-scaled and value-scaled modulation of choice. A finding that an SC perturbation spared detection, or that a value-circuit perturbation collapsed detection of large changes, would falsify the loop hypothesis as stated.

8 Discussion

The central result is architectural: for this class of recurrent attention agent, coupling between the policy and value pathways is a structural prerequisite for learning, on par with the choice of learning signal itself. A reasonable-looking modular design — two specialised recurrent streams cleanly separated for policy and value — does not merely underperform when the streams are isolated; it fails to acquire the task entirely. The margin between the coupled and uncoupled designs is among the largest one can produce by a single structural edit: a ~ 79 -point hit-rate gap and a collapse from a calibrated detector (accuracy 0.868, hit rate 0.794) to an agent that never once declares a change (0.000 hits across 500 episodes). Two jointly necessary conditions

bound the effect: the streams must share a substrate, and the image-grounded priority stream must command the decision.

8.1 A reinforcement-learning reading: credit assignment closes or opens the loop

The categorical nature of the failures is most naturally read through the actor–critic credit-assignment problem. In the coupled configuration the shared deep memory forces the critic to evaluate the *same* evolving state the actor acts from, and the actor’s own output writes that state. “Pressing now looks good” is then a statement about the state the actor actually occupies; credit lands where it was earned, and the actor can discover that declaring at change onset is rewarded. When the coupling is severed, the actor acts from a private memory while the critic values a different private memory; the two representations drift apart, and the advantage handed to the actor is computed on the wrong state. Credit is systematically misattributed, and the policy retreats to the only basin that is safe under any value estimate — never act, never be penalised — and stays there. The always-wait collapse is the rational endpoint of an agent that cannot tell when acting helps; it is a learning failure, not a capacity failure, since the network retains the parameters and pathways to express the task. The swapped-routing control fails for a complementary reason: value and policy are coupled, but the actor is now fed the value stream, whose residual is memory rather than image and which therefore carries almost no live visual content. However clean the credit assignment, there is nothing in the actor’s input that can see the change, so there is nothing to which credit can be assigned. The two controls thus dissociate the two necessary conditions cleanly.

8.2 Mapping onto the primate attention and value systems

The working loop has a natural counterpart in the biology. A grounded priority and commitment system — superior colliculus together with the visual cells of the frontal eye field — commits the action and is causally necessary for covert selection: SC inactivation produces a large covert-attention deficit while cortical gain is left intact (Zénon and Krauzlis, 2012; Lovejoy and Krauzlis, 2010), is more devastating than FEF inactivation (Bollimunta et al., 2018), and acts as the threshold that converts cortical evidence accumulation into a committed response (Stine et al., 2023); caudate attentional modulation itself depends on SC, implicating a colliculo-striatal–cortical loop (Herman et al., 2020). During covert attention FEF carries the signal on its image-grounded visual cells while movement cells are suppressed (Thompson et al., 2005; Moore and Fallah, 2001, 2004), and the decision requires grounded sensory evidence accumulated to a bound (Gold and Shadlen, 2007). The single closest parallel to the swapped-routing failure is the finding that inactivating sensory cortex impairs discrimination while inactivating the downstream accumulator does not (Katz et al., 2016): remove the grounded representation and the decision fails, exactly as when the actor is fed the memory stream. The value side maps onto the actor–critic organisation of the basal ganglia (O’Doherty et al., 2004; Maia, 2009), with the loop closed by a dopaminergic reward-prediction-error teaching signal from SNc (Schultz et al., 1997; Hollerman and Schultz, 1998) acting through a corticostriatal three-factor plasticity rule (Morita and Kato, 2014). That value and decision signals can ride a single substrate is itself observed in LIP (Platt and Glimcher, 1999; Sugrue et al., 2004), supporting the shared-memory premise. The superior colliculus is best understood here as a selective-attention and commitment node rather than an evaluator (Krauzlis et al., 2018).

8.3 Honesty about the mapping

The correspondence is conceptual motivation, not a literal isomorphism, and its limits must be stated plainly. First, and most importantly, the substrate and direction of the critic-to-

actor coupling differ between model and brain. In the model the coupling is a *shared recurrent working memory* that the actor writes and both streams read; in the brain the value-to-actor influence is a *broadcast dopamine scalar* flowing from the critic system to the actor, not a shared writable state. Both a write direction (priority $\rightarrow$ shared memory) and a teaching direction (critic gradient $\rightarrow$ actor) exist in the model, so it is genuinely a closed loop; but the brain analogue should be read as the dopaminergic prediction-error precedent for “learning requires the loop,” not as an architectural match. Second, the dorsal/ventral actor–critic split is itself contested — action-value signals appear broadly across striatum — so we present it as an organising idealisation rather than settled anatomy (Maia, 2009). Third, the requirement that the grounded stream command the action is a moderate and partly contested parallel: reward and prior information can in some studies enter the sensory or drift-rate pathway directly, so the dissociation is not absolute. Fourth, the priority and value streams do *not* map onto two distinct attention nodes: the priority stream resembles the FEF-visual and SC commitment integrator, while the value stream corresponds to the value/critic system of the basal ganglia and dopamine, which is not an attention node. LIP, FEF and SC are each a single priority map, not an actor/critic pair.

8.4 Novelty and relation to prior models

We are not aware of a prior model that splits a recurrent vision encoder into priority and value streams on a shared memory, requires that coupling for learning, or requires the grounded stream to be the actor. The closest machine-learning result on actor/critic representation sharing reaches the opposite conclusion for feedforward representations, where separating the actor and critic representations helps. The contrast is informative rather than contradictory: for a recurrent, image-grounded agent whose decision depends on reading a live image at the moment of action, the shared recurrent substrate is what allows credit to reach the actor, and coupling becomes required.

8.5 Limitations

The claims are bounded by their scope. The result is a controlled comparison within a single agent family on a single task family, and the brain mapping is an analogy whose direction and substrate differ from the biology as noted above. The collapses are categorical and reproducible across 500 fresh episodes per configuration, but the present numbers are from trained checkpoints under a stochastic policy rather than an exhaustive seed sweep, and we do not claim that no alternative uncoupled design could ever learn the task — only that the minimal, parameter-matched uncoupling we tested does not. The functional roles we assign to the two streams are inferred from architecture and behaviour and are made falsifiable below rather than asserted.

8.6 The network as a generator of falsifiable predictions

Because the working configuration is interpretable and manipulable, it yields concrete, testable predictions framed as a double dissociation. Clamping or lesioning the value (q) circuit should degrade value and context modulation — the cue-value and reliability effects — while sparing change detection; lesioning the priority (μ /SC) node should instead collapse detection itself. This maps directly onto SC inactivation abolishing covert detection versus a value-circuit manipulation that leaves detection intact (Katz et al., 2025; Krauzlis et al., 2018), and it is testable at three levels: behaviourally, the coupling-required and grounded-must-act conditions should hold at training time; representationally, change-evidence should decode preferentially from the priority stream and value/context from the value stream; and causally, perturbing priority attention should shift behaviour and decision threshold while perturbing value attention should

shift critic value with behaviour spared. A reward-trained network constrained to be interpretable thus serves not as a claim about the brain but as a generative source of mechanistic hypotheses for systems neuroscience.

9 Methods

9.1 Task

The environment is a spatially cued orientation-change-detection task with a horizon of $T = 29$ frames. Each $50 \times 50 \times 3$ frame is partitioned into a 10×10 grid of 5×5 patches ($N = 100$ tokens). Stimuli are oriented patches whose orientations are corrupted by frame-to-frame Gaussian noise ($\sigma = 5$) to set difficulty. A coloured cue is presented on a single early frame and then removed; its side indicates the likely change location and its colour sets the trial’s reward magnitude. The cue’s reliability (validity $\in \{0.25, 0.50, 0.75, 1.00\}$) is the probability that an upcoming change, present on half of trials, occurs at the cued location. On change trials the target orientation shifts by a magnitude of up to roughly 64° , with onset jittered across frames (change index in roughly $[11, 25]$) and the new orientation persisting to the end of the episode; the change magnitude is reduced over training by a performance-gated curriculum. On each frame the agent emits a binary wait/declare action: declaring at or after the change is a hit, declaring before it is a false alarm, and waiting out a no-change trial is a correct rejection. Reward is delivered for hits (scaled by the cue-colour value) and for correct rejections.

9.2 Agent

The agent is a conv-free recurrent vision transformer (Section 2). Each patch is embedded by a shared per-patch network into a $d = 128$ -dimensional token with learned positional and temporal codes; cross-location information flows only through attention. The encoder maintains two per-token recurrent memories, $\mathbf{H}_1$ (sensory) and $\mathbf{H}_2$ (deep), advanced by independent LSTM cells, and reads them through two single-head, pre-norm cross-attention streams—a priority stream (μ) feeding the actor and a value stream (q) feeding the critic—differing only in key/value bank and residual (Equations (2)–(3)). The actor and the distributional critic are one-dimensional-convolutional heads over the token axis; the actor emits a wait/declare logit pair and the critic a five-quantile distribution over returns. The three configurations of the central comparison—V1 (shared memory), V2 (independent), V3 (swapped routing)—hold the front-end, the heads, the parameter budget and the entire training recipe fixed, differing only in inter-stream coupling and readout routing.

9.3 Training

Training uses an offline actor–critic objective with identical hyperparameters and identical update counts across the three configurations. The actor loss is a maximum-a-posteriori policy-improvement (MPO/MAP) term blended with a behaviour-cloning term; the critic is trained with a quantile Bellman loss against a periodically hard-copied target network that supplies both the bootstrap target and the policy-improvement reference. Whole episodes are stored in a prioritised replay buffer with priorities set by per-episode quantile error, and a brief forced-wait warm-up trains the critic on full trajectories before the actor begins to act.

9.4 Evaluation

Each trained configuration is evaluated on 500 fresh episodes under a stochastic (sampled) policy and scored by signal-detection category (hit, miss, false alarm, correct rejection), from which we compute overall accuracy, hit rate on change trials, false-alarm rate on no-change trials, and

reaction time on hits. For the working configuration, the cueing analysis splits hit rate by cue reliability and by cued versus uncued change location, with per-validity cells of $n = 300$ and per-magnitude psychometric cells of $n = 250$; reaction time is summarised by its median within condition. Across all 500 episodes the two collapsed configurations declare a change exactly zero times, so their hit and false-alarm rates are both 0.000 and no psychometric or reaction-time structure is defined. The three configurations of the central comparison correspond to internal model identifiers `v11_part2` (V1, shared memory), `v11_part4` (V2, independent) and `v11_part3` (V3, swapped routing).

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